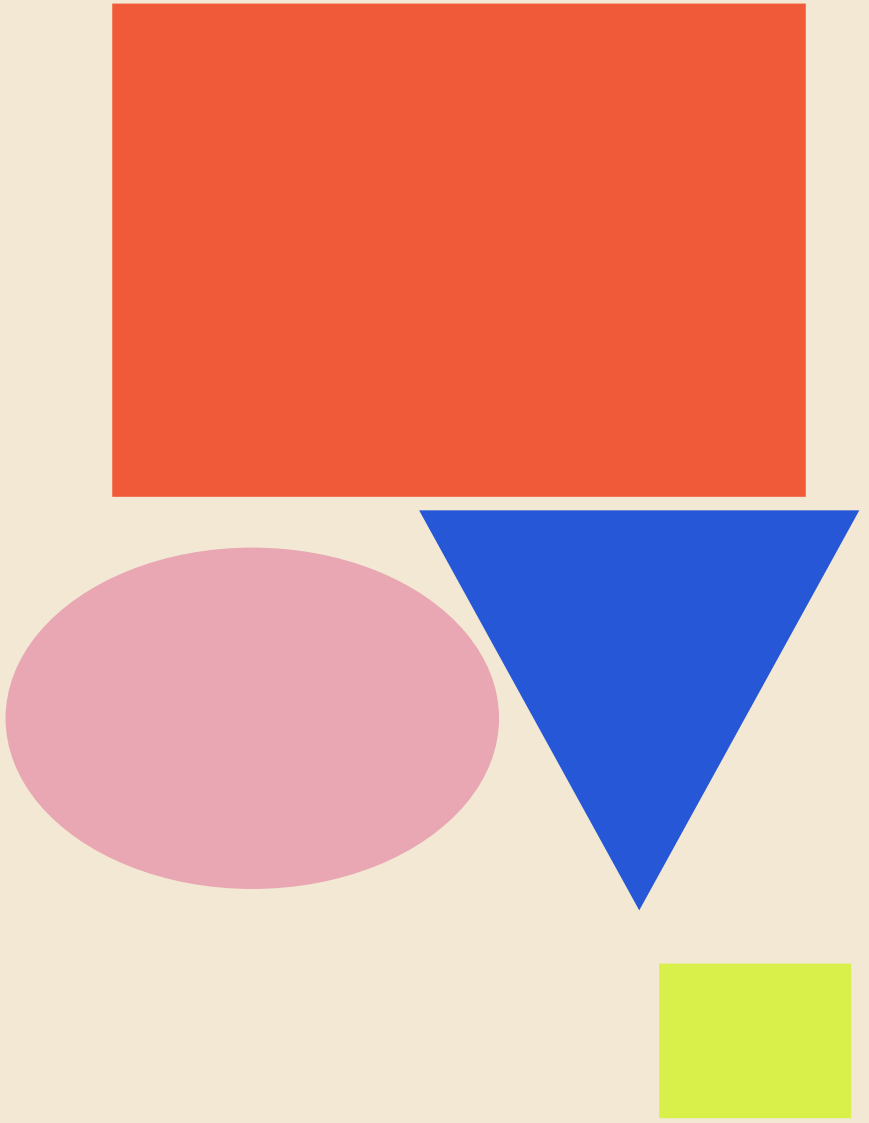
DATA SYSTEMS ADMINISTRATION · ENTRY-LEVEL COURSE

CYBER SECURITY FOUNDATIONS



From binary to passwords — five units that turn high-school graduates into junior system administrators.

Audience: recent high-school graduates · No prior IT experience needed

COURSE ROADMAP

FIVE STOPS TO YOUR FIRST ADMIN JOB

01 THE LANGUAGE OF COMPUTERS

Read binary and hex like a second language.

bits · bytes · ASCII · hex

02 IP FUNDAMENTALS

Address any device, and subnet by hand.

IPv4 · subnet masks · CIDR

03 CONNECTIVITY

Pick copper, fiber, or wireless for the job.

Ethernet · fiber · Wi-Fi

04 OPERATING SYSTEMS

Find your way around Linux and Windows.

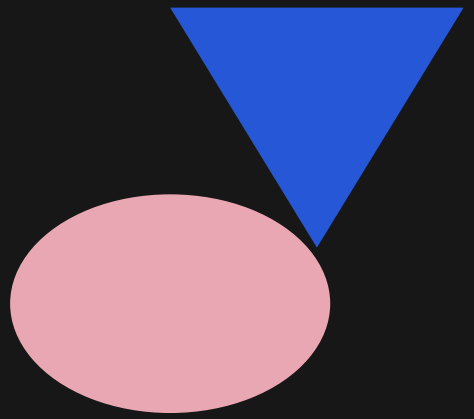
users · services · CLI

05 SECURITY FOUNDATIONS

Build password and MFA habits that stop attacks.

passphrases · MFA · phishing

UNIT 01



01

THE LANGUAGE OF COMPUTERS

Why everything a computer knows fits into 1s and 0s.

IN THIS UNIT bits & bytes · binary → decimal · ASCII & hex in the wild

COMPUTERS KNOW EXACTLY TWO THINGS

A computer is billions of microscopic switches. Each one is either **ON** or **OFF** — there is no "maybe".

We write ON as **1** and OFF as **0**. One switch is one **bit**. Eight bits make one **byte**.

Every photo, song, password, and video game is stored as a long pattern of these bits. That is the entire trick.

Anchor image: a row of light switches — the *pattern* of ons and offs is the message.

ONE BYTE = 8 BITS

1 0 1 0 1 1 0 0



■ switch ON (1) ■ switch OFF (0)

This exact byte stores the number **172** — next page shows how.

---

2 states per switch: **ON** and **OFF**. The whole digital world is built from this one fact.

Background: Brookshear & Brylow, Computer Science: An Overview, ch. 1 (2024)

READING BINARY: EACH POSITION DOUBLES

STEP 1 Write the place values: start at 1 on the right, double as you go left.

STEP 2 Put the bits underneath, one per column.

STEP 3 Add up only the places that have a 1.

TRY IT What number is **00001111**?

Answer at the bottom of the page.

value	128	64	32	16	8	4	2	1
bit	1	0	1	0	1	1	0	0
keep	128	—	32	—	8	4	—	—

128 + 32 + 8 + 4 = 172

REMEMBER The biggest byte is **11111111 = 255.** That is why every part of an IP address maxes out at 255 — Unit 02 uses this constantly.

TRY IT answer: 00001111 = 8 + 4 + 2 + 1 = 15 · Conversion method: Duke & Obidunnu (2009)

LETTERS ARE JUST NUMBERS IN COSTUME

ASCII — THE DRESS-UP CODE

Every character on your keyboard has an agreed number, so any computer can store text as bytes.

"A" = 65 · "a" = 97 · "0" = 48 · space = 32

So the word **"Hi"** is really two bytes: 72 then 105 — the pattern 01001000 01101001 on disk.

HEX — BINARY'S SHORT-HAND

Hexadecimal counts 0–9 then **A–F** (A = 10 … F = 15). One hex digit covers 4 bits, so **one byte = two hex digits**.

Our number **172** becomes **AC**: A = 10 sixteens (160) + C = 12.

Admins read hex daily — it is binary, without the eye strain.

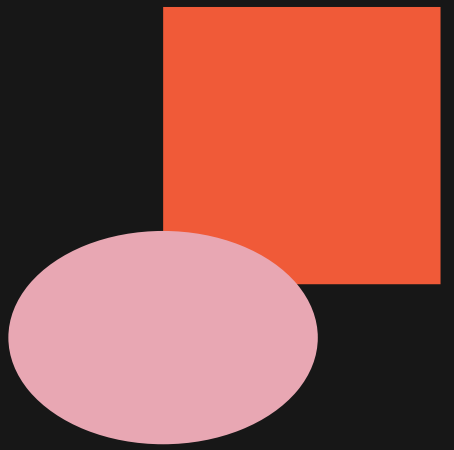
WHERE YOU WILL ACTUALLY SEE THIS

MAC addresses — every network card's ID: **7C:2E:BD:41:0A:9F** (pure hex)

Error codes — Windows **0x80070005** means "access denied" · Logs & configs — colors, memory offsets, permission bits

Background: ASCII standard (1963); Brookshear & Brylow (2024), ch. 1 — data representation

UNIT 02



02

IP FUNDAMENTALS

Give every device an address, and split networks with confidence.

IN THIS UNIT dotted decimal · subnet masks & CIDR · hands-on subnetting drills

AN IP ADDRESS IS A STREET ADDRESS

192.168.1.10

NETWORK = the street you live on

HOST = your house number

Each of the four numbers is an **octet** — one byte, so **0–255** (Unit 01 payoff!). The **subnet mask** is the ruler that says where "street" ends and "house" begins.

FOUR THINGS EVERY DEVICE NEEDS TO GET ONLINE

1. **IP address** — who you are on the network
2. **Subnet mask** — which network you belong to
3. **Default gateway** — the door out (your router)
4. **DNS server** — the phone book that turns names into numbers

Source: Postel, RFC 791 — Internet Protocol (1981), addressing section

SUBNETTING: ONE NETWORK BECOMES FOUR

You start with **192.168.1.0/24** — 256 addresses. "Borrow" 2 bits for the network and you get a **/26**: four subnets of 64 addresses each (62 usable hosts; the first and last addresses are reserved).

SUBNET	NETWORK	FIRST HOST	LAST HOST	BROADCAST
1	192.168.1.0	.1	.62	.63
2	192.168.1.64	.65	.126	.127
3	192.168.1.128	.129	.190	.191
4	192.168.1.192	.193	.254	.255

HOW TO DERIVE IT

1. New mask /26 → mask's last octet is 192.

2. Block size = 256 − 192 = **64**.

3. Count networks up by 64: 0, 64, 128, 192.

4. First host = network + 1; broadcast = next network − 1; last host = broadcast − 1.

LAB DRILL Derive the same table for a /27 (block size 32) — then verify it in Cisco Packet Tracer.

Reference: RFC 4632 — CIDR (2006); taught by drilling worked examples

SOME ADDRESSES STAY HOME

PRIVATE RANGES — FREE FOR EVERYONE

10.0.0.0 – 10.255.255.255

172.16.0.0 – 172.31.255.255

192.168.0.0 – 192.168.255.255

Anyone may use these indoors; they never appear on the public internet. Your router shares **one public IP** for the whole house — a trick called **NAT**.

IPV6 — THE BIGGER PHONE BOOK

IPv4 gives ~4.3 billion addresses — phones, laptops, watches, and doorbells used them all up.

IPv6 stretches to **128 bits**, written in hex:

2001:db8::1

Same street-address idea — just a much longer street.

THEN vs NOW — READ THE ACTUAL SPECS

RFC 791 (1981) — the original IPv4 spec; the addressing section is only a few pages and surprisingly readable.

RFC 2460 (1998) — IPv6; compare them side by side and watch one idea grow up.

Sources: RFC 1918 — private ranges (1996); RFC 791 (1981); RFC 2460 (1998)

UNIT 03

03

CONNECTIVITY

Copper, fiber, or air: pick the right path for the data.

IN THIS UNIT Ethernet over copper · fiber optics · Wi-Fi & radio basics

ETHERNET — FROM 10 Mbps TO 400,000

1983	1995	1999	2002	2010	2017
10 Mbps	100 Mbps	1 Gbps	10 Gbps	40/100 Gbps	400 Gbps

COPPER IN YOUR HANDS TODAY

The cable at your desk is **twisted-pair copper** — Cat5e or Cat6 — ending in the chunky **RJ45** plug.

One rule never changes: **100 meters max per run**. Longer than that and the signal fades.

"Ethernet" is just the IEEE **802.3** working group's name for the standard.

WATCH OUT FOR

Interference — copper is an antenna; keep it away from power cables and motors.

Distance — past 100 m you need a switch, or a different medium.

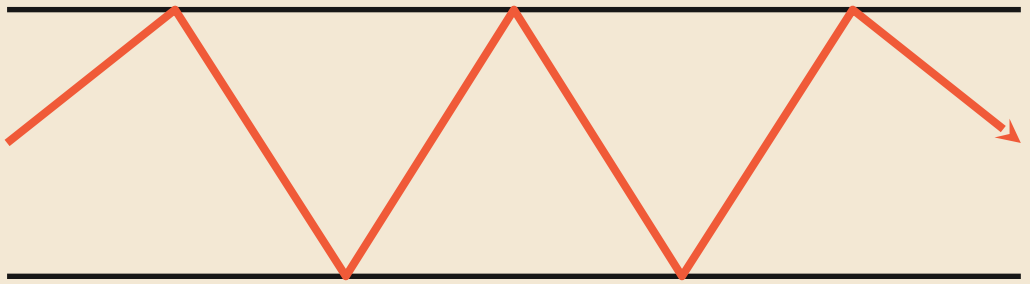
Cheap cable — a bad crimp causes the weirdest network bugs you will ever chase.

Source: Law et al., Evolution of Ethernet Standards in IEEE 802.3 (2013)

DATA AT THE SPEED OF LIGHT — LITERALLY

A fiber cable is a hair-thin strand of glass. Light goes in one end and **bounces off the walls** all the way down — a physics trick called **total internal reflection**.

Flashing the light on and off is just bits again: **light = 1, dark = 0**.



Light zigzags inside the glass core, trapped by total internal reflection.

SINGLE-MODE — THE MARATHONER

One thin core, one straight beam. Carries data **kilometers** — under-city links, ISP backbones, between buildings.

MULTI-MODE — THE SPRINTER

A wider core, many bouncing beams. Cheaper gear, good for **hundreds of meters** — inside one building or data center.

Why admins love it: immune to electrical interference · insanely fast · this is what "fiber to the home" (FTTH) actually is.

Ghatak & Thyagarajan (1998)

WI-FI IS JUST RADIO, POLITELY SHARING

Wi-Fi (IEEE 802.11) sends bits through the air as radio waves. The catch: everybody shares the same air.

2.4 GHz

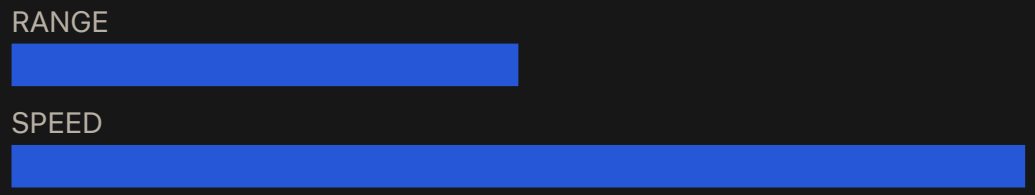
the old faithful



Reaches far and through walls — but crowded: neighbors, Bluetooth, even microwaves live here.

5 / 6 GHz

the fast lane



Much faster with more room — but walls eat it. Same trade-off, every time.

WHAT KILLS YOUR SIGNAL

Walls & metal — every wall costs you bars · **Channel crowding** — on 2.4 GHz only channels **1, 6, 11** don't overlap · **Distance** — radio fades fast indoors

Background: Stallings, Wireless Communications & Networks (2009), Wi-Fi chapter

IF IT DOESN'T MOVE, WIRE IT

01 COPPER — the default

Cheap, easy, everywhere. Desks, printers, wall jacks.

≤ 100 m · RJ45 · Cat5e/6

02 FIBER — the distance king

Fastest and farthest. Between buildings, server uplinks, ISP links.

100s of m – km · immune to noise

03 WIRELESS — the freedom option

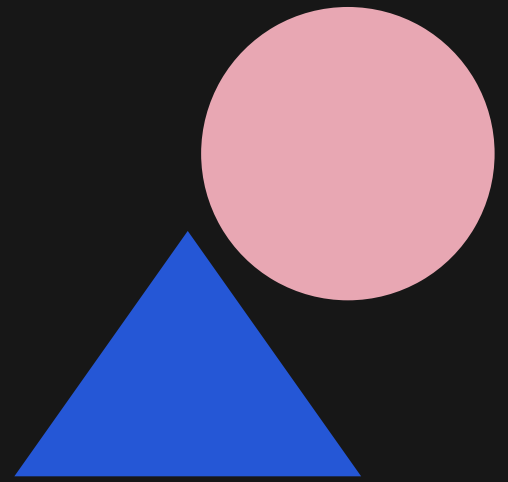
For anything that moves: phones, laptops, guests, IoT gadgets.

last 10–30 m · shared air

RULE OF THUMB Desktops and servers → cable. Buildings and backbones → fiber. People and phones → Wi-Fi.

Standards: IEEE 802.3 (Ethernet) · IEEE 802.11 (Wi-Fi)

UNIT 04



04

OPERATING SYSTEMS

Linux runs the servers, Windows runs the desks — you will run both.

IN THIS UNIT what an OS actually does · users & permissions · command line basics

THE OS IS THE BUILDING MANAGER

Hardware is the building; apps are the tenants. The **operating system** decides who gets what, and keeps the peace.

Files — where data lives

Users — who is allowed to do what

Memory & CPU — which program gets attention

Network — how data leaves the machine

LINUX

Free & open source · runs most of the internet's servers and all of the cloud's favorites · you will type commands in a **terminal** · everything is a file.

WINDOWS

Runs most office desktops · graphical first, **PowerShell** when you mean business · **Active Directory** manages users across a whole company.

Same jobs, two accents — a good admin is bilingual. Source: Tanenbaum & Bos, Modern Operating Systems (2015)

THREE JOBS EVERY ADMIN DOES DAILY

01 Manage users — give least privilege

Every account gets *only* the access its job needs. Linux: permissions & sudo · Windows: user groups & Active Directory.

02 Find your way — from the command line

Clicking is fine once; typing is repeatable. Linux terminal: **pwd · ls · cd** · Windows PowerShell: **dir · ipconfig**.

03 Keep it patched — updates are armor

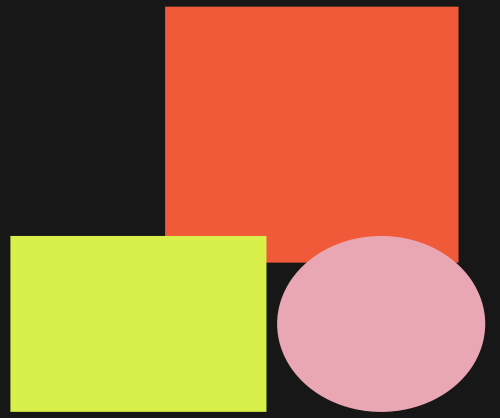
Most real-world hacks exploit bugs that already have a fix. An updated system is a boring target.

YOUR FIRST ADMIN COMMAND

$ ping 192.168.1.1 → "reply from..." means the road to your router is alive. No reply? Start checking cables and Wi-Fi.

Practice scenario — try it in the lab VM. Background: Silberschatz et al., Operating System Concepts (2006)

UNIT 05



05 SECURITY FOUNDATIONS

The easiest attacks target people, not machines.

IN THIS UNIT passphrases that work · multi-factor authentication · spotting phishing

LENGTH BEATS COMPLEXITY

Tr0ub4dor&3

11 characters, hard for *you* to remember — but short, so a computer chews through it fast.

coffee-river-paper-moon

23 characters of plain words — easy to remember, brutally hard to guess.
Bars show relative guessing effort (illustrative).

THE MODERN RULEBOOK (NIST)

- Go **long** — passphrases beat P@ssw0rd!
- **Unique** per account — reuse is how one leak becomes ten
- Let a **password manager** remember for you
- No forced monthly changes — they make people weaker, not safer

MEMORY TRICK Pick 3–4 random words — a song line, three unrelated nouns — and string them together. Your brain keeps the story; the length does the protecting.

Sources: NIST SP 800-63B Digital Identity Guidelines; Bonneau & Shutova (2012) on passphrases

ONE LOCK IS NOT ENOUGH

KNOW
something you know — password, PIN, passphrase

+

HAVE
something you have — phone, app code, security key

+

ARE
something you are — fingerprint, face

Any **two** of the three = multi-factor authentication. A leaked password alone gets an attacker nowhere.

SPOT THE PHISH — FIVE RED FLAGS

1. **Urgency** — "your account closes in 24 hours!"
2. **Almost-right sender** — micr0soft.com, paypa1.com
3. **Weird links** — hover first; the real address shows at the bottom
4. **Surprise attachments** — "invoice_2084.pdf.exe"
5. **Asks for secrets** — real IT *never* asks for your password or MFA code

Sources: Ometov et al. (2018) — MFA survey; Salahdine & Kaabouch (2019) — social engineering survey

YOUR FIRST-WEEK CHECKLIST

- ☐ **Turn on automatic updates** — on every machine you touch, including your phone.
- ☐ **Lock every screen** — walk away = lock it. Win+L or Ctrl+Cmd+Q until it's a reflex.
- ☐ **Start a password manager + MFA** — protect your email first; it resets everything else.
- ☐ **Hover before you click** — check the sender and the link, every single time.
- ☐ **Ask "do they need this access?"** — least privilege for every account you create.
- ☐ **Back up before you change anything** — the undo button of real life.

WHERE THIS LEADS (NICE FRAMEWORK)

IT support → system / network administrator → security analyst. This course is rung one.

Source: Newhouse et al., NICE Cybersecurity Workforce Framework (2017)

WRAP-UP

YOU SPEAK MACHINE NOW

01 Computers are switches — every file is a pattern of bits.

02 An IP is a street address; masks decide who your neighbors are.

03 Copper for desks, fiber for distance, Wi-Fi for freedom.

04 The OS is the building manager — Linux and Windows, same jobs.

05 Long passphrases + MFA + a skeptical eye stop most attacks.

KEEP PRACTICING

· **Cisco Packet Tracer** — build the subnets from Unit 02

· **Wireshark** — watch real packets fly by

· **A Linux VM** — break it, fix it, repeat

· **RFC 791** — read the original spec, it's short

COURSE SOURCES — RFC 791 (IPv4) · RFC 1918 (private addressing) · RFC 4632 (CIDR) · IEEE 802.3 / 802.11 · NIST SP 800-63B (digital identity) · Tanenbaum & Bos, Modern Operating Systems (2015) · Brookshear & Brylow, Computer Science: An Overview (2024) · Law et al. (2013) · Salahdine & Kaabouch (2019) · Ometov et al. (2018)